

An AI Literacy Curriculum Framework

Scientific Reasoning with Lunar Thin Sections and LLMs

The Moon is my beautiful impossibility.

Ping Wang

April 2026

Tutorial • Colab Notebooks • Lunar Petrographic Exhibit

An AI Literacy Curriculum Built on Lunar Thin Sections

Program Vision

This AI literacy curriculum is founded on the recognition that future scientific discovery will increasingly depend on the ability to integrate vast and complex datasets, synthesize knowledge across disciplines, and reason effectively from multiple lines of evidence. Accordingly, this curriculum focuses on exploring the integration of Large Language Models (LLMs) into scientific reasoning workflows for lunar petrography, petrogenesis, and planetary history reconstruction. Students will examine how LLMs can assist in interpreting lunar thin sections, inferring petrogenetic histories, synthesizing geochemical and mineralogical evidence, and reconstructing the geological evolution of the Moon, while critically assessing the capabilities and limitations of AI systems.

Contents

- Unit 1: Lunar Thin Sections
- Unit 2: LLMs and Their Outputs
- Unit 3: Constructing evidence-based Inferences from Lunar Rocks
- Unit 4: AI in Science: Capabilities & Critical Limitations
- Unit 5: A Lunar Petrographic Exhibit

Learning Outcomes

By the end of the curriculum, learners will be able to:

- Make and record systematic petrographic observations and clearly distinguish observation from interpretation.
- Explain how an LLM generates text.
- Verify an LLM-generated scientific claim against evidence.
- Evaluate a chain of scientific reasoning for unsupported links and assumptions.
- Assess capabilities, limitations, biases, and hallucinations in AI reasoning.
- Write evidence-based petrographic reports that support interpretations of lunar history.
- Design a Lunar petrographic exhibit through artistic expression.

Advanced Research

This curriculum utilizes a multi-level engagement strategy. For those seeking greater challenges, we provide individualized mentoring and research support, enabling them to pursue advanced projects, including: (1) developing LunarBench, a benchmark for evaluating scientific reasoning in lunar petrography, (2) building a lunar knowledge corpus, which organizes returned lunar sample descriptions, lunar atlases, petrographic reports, and scientific publications into a machine-readable resource for AI-assisted lunar science, (3) developing an agentic RAG system that assists with scientific investigation and hypothesis generation, and (4) creating an lunar petrographic exhibit with AI-assisted narrative.

Audience and Prerequisites

This curriculum is written for advanced secondary or lower-division undergrad students and assumes no prior geology as a compressed multi-day workshop. No prior knowledge of machine learning is assumed either. Unit 2 supplies exactly as much LLM mechanism as a young scientist needs to reason about reliability, and no more.

Resources

An instructor-curated image pack, a tutorial, hands-on colab notebooks, and LLM access. With a multimodal LLM model, learners can upload a thin section image and watch the model describe what it sees. With a text-only model, learners feed the model their own written observations, which usefully isolates the model's role as an inference-from-description rather than perception.